

First year Robotics and AI student at UCL with experience in full stack development, embedded systems and production applications. Proficient in Python, C#, Javascript and React with real life experience in hardware control, networking, game development and cloud services. Interested in software engineering, Artificial Intelligence and robotics.

EDUCATION

UCL	London/United Kingdom
Robotics and AI	2026 - 2030
WBGS	Watford/United Kingdom
Achieved A*A*AA at A levels	2024 - 2026
Achieved 9887777766 at GCSE	2019 - 2024

PROJECTS

UAV Hangar | Python

- Collaborated in a team to create a fully automatic UAV hangar
- Established secure communications between two Raspberry Pi's connected with TCP over SSL for tag-based aircraft detection
- Integrated two Stepper Motors with a pulley system and pressure detection for automatic physical movement

Full Stack website | Javascript, React

- Co-Led developing a full stack website connecting tutors and tutees
- Implemented a firebase authentication and database for scaling
- Integrated Stripe into the node.js backend to support secure payment
- Developed a seamless, clean and responsive UI for accessibility and ease of bookings

Tower Defence Game | C#, Godot

- Developed a 2D Tower Defence game with a flashcard based revision system
- Implemented wave-based enemy spawning, multiple enemy and tower types, custom projectile and tower upgrade mechanics
- Designed a modular architecture separating enemy, tower and question management systems

WORK EXPERIENCE

Benham and Reeves | work experience | London 2024

- Identified an opportunity to streamline workflow in accounts department with a password management system
- Developed in python and later handed off to the IT department for further development and integration
- Found out user requirements and adapted a solution for the accounts department

EXTRA CURRICULAR ACTIVITIES

- Piano Grade 6
- Engineering Society - Lead, WBGS
- Computer Science Society - Student Lead, WBGS
- Hockey Team - WBGS

AWARDS AND CERTIFICATIONS

- Industrial Cadets Gold
- International Astronomy and Astrophysics Competition - Bronze honours
- International Computer Science Competition - Finalist
- C# Foundational Certification

TECHNICAL SKILLS

Languages - Python, C#, JavaScript

Frameworks and Tools - Git, Godot, React

Cloud and Backend - Firebase, Cloud, Node.js

Hardware and Networking - Raspberry Pi, GPIO, TCP/IP, SSL, BlueZ